eCH-1234 Template Quarto Document

September 3, 2026

Abstract Brief summary of the purpose of the document.

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Note

This document uses a gender-neutral formulation when referring to persons. This is based on the guidelines (German) of the Federal Chancellery. Depending on the situation, paired forms (citizens), gender-abstract forms (insured person), gender-neutral forms (insured person) or para-phrases with-out personal reference are used. The generic masculine (citizen) is not permitted. Full forms are used in continuous texts, i.e. in texts consisting of formulated sentences. Short forms can be used in ab-breviated text passages, namely in tables. The short form is used with a slash but without an ellipsis (referent). Gender asterisks and similar spellings are not used.

1 Introduction

1.1 Status

Approved: This document was approved by the Experts' Committee. It has normative power for the defined field of application in the determined scope of application.

1.2 Scope of application

The information in this chapter should provide the reader with a brief overview of what this standard is intended for. Information about the following matters may be helpful here.

1.3 We can have sub-headings

And write some text.

1.3.1 Also sub-sub-headings

And write some more text. Maybe even with a pretty image.

Figure 1: Always add some text to describe what the image shows.

When displaying diagrams, try to write them in Mermaid JS straight away; this makes changes in the future or translations straightforward.

2 Technical notes

We use the ROBOT CLI in our project (Jackson et al. 2019), especially for it's ability to run the HermiT reasoner (Glimm et al. 2014).

3 Data Model

3.1 Genre

A musical category in the Chinook dataset.

Target Class: `:Genre`

Table 1: properties Genre

Description	Path	Type	Cardinality
Name	`schema:name`		1..1
Part of	`schema:partOf`	`:Genre` or `sh:IRI`	0..*

3.2 Invoice

A purchase receipt in the Chinook dataset.

Target Class: `schema:Invoice`

Table 2: properties Invoice

Description	Path	Type	Cardinality
Customer: An invoice must be linked to exactly one customer.	`schema:customer`	`schema:Person`	1..1
Total payment due: An invoice must define a total payment due as a schema:QuantitativeValue.	`schema:totalPaymentDue`	`schema:QuantitativeValue`	1..1
Has part: An invoice must have at least one line item (OrderItem).	`schema:hasPart`	`schema:OrderItem`	1..*
Date created	`schema:dateCreated`	`xsd:date`	0..*

3.3 Music album

A collection of tracks in the Chinook dataset.

Target Class: `schema:MusicAlbum`

Table 3: properties Music album

Description	Path	Type	Cardinality
Name: Every album must have a name.	`schema:name`	`xsd:string`	1..1
Artist: Person or music group who created the album.	`schema:byArtist`	`schema:Person` or `schema:MusicGroup`	0..*

3.4 Music Recording

A single music track in the Chinook dataset.

Target Class: `schema:MusicRecording`

Table 4: properties Music Recording

Description	Path	Type	Cardinality
Name: Every track must have a name.	`schema:name`	`xsd:string`	1..1
In Album: A track can only belong to a valid schema:MusicAlbum.	`schema:inAlbum`	`schema:MusicAlbum`	0..*
Author: The person or group who wrote the track.	`schema:author`		0..*
Genre	`schema:genre`	`:Genre`	1..1
Duration: Track duration must be expressed as a schema:QuantitativeValue.	`schema:duration`	`schema:QuantitativeValue`	0..*
Content Size	`schema:contentSize`	`schema:QuantitativeValue`	0..*

3.5 Organisation

A company or organization in the Chinook dataset.

Target Class: `schema:Organisation`

Table 5: properties Organisation

Description	Path	Type	Cardinality
Name: Every organisation must have a name.	`schema:name`	`xsd:string`	1..1

3.6 Person

Any person (employee, customer, or custom person) present in the dataset.

Target Class: `schema:Person`

Table 6: properties Person

Description	Path	Type	Cardinality
Given Name: Every person must have a given name.	`schema:givenName`	`xsd:string`	1..1
Family Name: Every person must have a family name.	`schema:familyName`	`xsd:string`	1..1
Email Address: If an email is provided, it must follow a standard email format.	`schema:email`	`xsd:string`	0..*
Birth date: A person should have a valid birth date.	`schema:birthDate`	`xsd:date`	0..*
Address	`schema:address`	`schema:PostalAddress`	0..*
Works for: An employee can report to another person.	`schema:worksFor`	`schema:Person` or `schema:Organization`	0..*
Job title	`schema:jobTitle`		0..1
knows	`schema:knows`	`schema:Person`	0..*

3.7 Quantitative Value

A numerical value with an associated unit.

Target Class: `schema:QuantitativeValue`

Table 7: properties Quantitative Value

Description	Path	Type	Cardinality
Value: A quantitative value must have exactly one numeric value.	`schema:value`		1..1
Unit Code: A quantitative value must specify its unit via a unitCode URI.	`schema:unitCode`	`sh:IRI`	1..1

4 Data Retrieval

The master and reference data underlying this document are available as *Linked Data*.

The technological basis for this is the Resource Description Framework (RDF, Cyganiak et al. 2014), a central standard of the World Wide Web Consortium (W3C) for modeling data structures on the web. In RDF, information is not represented in classic tables, but as interconnected graphs. Each statement consists of a so-called triple (subject, predicate, object). This structure enables a machine-readable, interoperable, and cross-system unambiguous description of resources and their relations to one another.

For the storage and publication of this RDF data, LINDAS (Linked Data Service) is used, the official Linked Data service of the Swiss Federal Administration. LINDAS functions as a so-called *triple store*, a specialized graph database optimized for the efficient storage and querying of RDF triples, which makes the data publicly available via a standardized interface.

The following chapter provides minimal instructions on how the data can be queried and retrieved from LINDAS.

```
BASE <https://example.org/>
PREFIX schema: <http://schema.org/>
SELECT *
WHERE {
    ?genre a <Genre> ;
        schema:name ?name .
}
LIMIT 10
```

The underlying data itself is maintained on GitHub as Turtle files.

```
@base <http://example.org/> .
@prefix genre: <http://example.org/genre/> .
@prefix schema: <http://schema.org/> .

genre:1 a <Genre> ;
    schema:name "Rock" .

genre:2 a <Genre> ;
    schema:name "Jazz" .
```

```
genre:3 a <Genre> ;
    schema:name "Metal" ;
    schema:partOf genre:1 .
```

5 Safety considerations

Information about the explicitly relevant legal bases or a note that during the implementation the rele-vant legal bases must be observed.

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8 Annex A - References

Cyganiak, Richard, David Wood, and Markus Lanthaler. 2014. *RDF 1.1 Concepts and Abstract Syntax*. W3C Recommendation. World Wide Web Consortium (W3C). https://www.w3.org/TR/rdf11-concepts/.

Glimm, Birte, Ian Horrocks, Boris Motik, Giorgos Stoilos, and Zhe Wang. 2014. “HermiT: An OWL 2 Reasoner.” *Journal of Automated Reasoning* 53 (3): 245–69.

Jackson, Rebecca C, James P Balhoff, Eric Douglass, Nomi L Harris, Christopher J Mungall, and James A Overton. 2019. “ROBOT: A Tool for Automating Ontology Workflows.” *BMC Bioinformatics* 20 (1): 407. https://doi.org/10.1186/s12859-019-3002-3.

9 Annex B - Cooperation and Verification

10 Annex C - Abbreviations and Glossary

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12 Annex E - Table of figures

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